THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 7 & 8: **Measurement of Time and Motion, Life Processes in Animals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 60 km/h	speed = distance ÷ time
2	**(b)** 60 km/h	speed = distance ÷ time
3	**(d)** 100 km	distance = speed × time
4	**(c)** 3 hours	time = distance ÷ speed
5	**(c)** 20 m/s	1 km/h = 5/18 m/s
6	**(b)** starch	partial digestion in mouth
7	**(c)** fats	breaks fats into droplets
8	**(b)** small intestine	inner lining
9	**(d)** large intestine to function properly	easier egestion
10	**(a)** alimentary canal	digestive tract

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	2 s (2).	2
12	54 km/h = 15 m/s so 20 m/s is faster (2).	2
13	Glucose + Oxygen → Carbon dioxide + Water + Energy (2).	2
14	Tiny air sacs in the lungs (1); exchange gases with blood (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Pendulum oscillates with fixed period (1.5); gears count the oscillations (1.5).	1½+1½
16	Gills cannot use air properly (3).	3
17	18 km/h (3).	3
	OR: 200/50 = 4 m/s (3).	3
18	Food digested and absorbed (3).	3
	OR: Pranayama and Tummo (1.5); improve health (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) s = d/t (1). (ii) 80 km/h (2). (iii) 6 h (2).	1+2+2
	OR: (i) No (1). (ii) 5 m/s (1). (iii) 6 m/s (1). (iv) 5.33 m/s (2).	1+1+1+2
20	(i) Lung damage (2). (ii) Inhaling others' smoke (1). (iii) Smoke-free areas, awareness (2).	2+1+2
	OR: (i) Lungs (1). (ii) Diaphragm (1). (iii) Balloons inflate (1). (iv) Inhalation (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 54 km/h	multiply by 18/5
2	**(a)** time period	T
3	**(c)** sundial	Sun's shadow
4	**(c)** metre per second	m/s
5	**(b)** speedometer	shows km/h
6	**(d)** gluten	damages the small intestine
7	**(a)** windpipe	trachea
8	**(a)** down and inwards	space reduces
9	**(d)** carbon dioxide	test for CO_2
10	**(d)** skin	also under water

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The total distance travelled by a vehicle (2).	2
12	Sundial; water clock (2).	2
13	Moves downward and ribs move up and out, so air enters (2).	2
14	Through gills that absorb dissolved oxygen from water (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Speed changes so we use total distance ÷ total time (3).	3
16	Thin walls with blood vessels (3).	3
17	Measures short intervals (1.5); athletics (1.5).	1½+1½
	OR: Water drips at a steady rate (1.5); marks show the time (1.5).	1½+1½
18	Neutralises acid, breaks fats (3).	3
	OR: Separate pathways (1.5); rings keep it open (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Bob on a thread (1). (ii) Length (2). (iii) Clocks (2).	1+2+2
	OR: (i) Change of position (1). (ii) Linear, circular, periodic (2). (iii) Speedometer (2).	1+2+2
20	(i) Mouth to anus (2). (ii) Functions (3).	2+3
	OR: (i) Breaks glucose (2). (ii) Walking, thinking (1). (iii) CO_2 and water (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET C

Chapters 7 & 8: **Measurement of Time and Motion, Life Processes in Animals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 50 km/h	speed = distance ÷ time
2	**(a)** 90 km/h	speed = distance ÷ time
3	**(d)** 60 km	distance = speed × time
4	**(a)** 3 hours	time = distance ÷ speed
5	**(b)** 15 m/s	1 km/h = 5/18 m/s
6	**(b)** oesophagus	carries food to stomach
7	**(a)** basic in nature	neutralises acid
8	**(c)** water and some salts	forms semi-solid waste
9	**(c)** rumination	partially digested food returns to mouth
10	**(d)** Liver	stored in gall bladder

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distance = speed × time (2).	2
12	Works at all times (2).	2
13	Through spiracles connected to tubes (2).	2
14	Damages lungs and can cause cancer (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Moves to and fro with regular time (3).	3
16	Carries nutrients and gases (3).	3
17	300 km (3).	3
	OR: 20 m/s (3).	3
18	Supply helpful bacteria (3).	3
	OR: Reaction to gluten harming the intestine (1.5); millets (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) A (1). (ii) 72 km and 54 km (2). (iii) A (1). (iv) 18 km (1).	1+2+1+1
	OR: (i) Using Earth's rotation and revolution (2). (ii) One rotation (1). (iii) One revolution around the Sun (2).	2+1+2
20	(i) More energy needed (1). (ii) Lungs, heart (1). (iii) Carries oxygen (1). (iv) Moves for faster breathing (2).	1+1+1+2
	OR: (i) Single vs multichambered (2). (ii) Rumen, rumination (2). (iii) For plant diet (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 72 km/h	multiply by 18/5
2	**(b)** oscillation	complete cycle
3	**(a)** at night or on cloudy days	needs sunlight
4	**(d)** linear motion	e.g. a car on a straight road
5	**(b)** odometer	total distance
6	**(d)** breathing	physical process
7	**(a)** alveoli	gas exchange
8	**(c)** lungs	model
9	**(b)** gills	dissolved oxygen
10	**(d)** circulatory system	heart and blood

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Motion along a straight line (1); a train on a straight track (1).	1+1
12	330 minutes (2).	2
13	Absorbs water and salts and forms stool (2).	2
14	Exhaled air turns lime water milky faster (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	36 km/h = 10 m/s so 11 m/s is faster (3).	3
16	Good digestion (3).	3
17	Speed and distance reading (3).	3
	OR: Horizontal line for rest (1.5); sloping line for uniform motion (1.5).	1½+1½
18	Protect delicate organs (3).	3
	OR: Large rumen stores grass (1.5); returned for chewing (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Travel planning, sports (2). (ii) Example computation (2). (iii) Consistency (1).	2+2+1
	OR: (i) Rising, flat, rising (3). (ii) Motion, rest, motion (2).	3+2
20	(i) Gills (2). (ii) Absorb oxygen (2). (iii) Frog (1).	2+2+1
	OR: (i) Chew well, eat on time, fibre (2). (ii) Better absorption (2). (iii) Smooth working (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.